

DBMS – WEEK 4

VILVAH

Orders and Order_Details Management

1. Objective

1. To design the **Orders** and **Order_Details** tables.
2. To manage customer product orders in the Vilvah database.
3. To store order date, quantity, unit price, discount, and total amount.
4. To perform order insertion and modification operations.
5. To generate customer order history and customer-wise order summary reports.

2. Database Design

2.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	NUMBER(10)	Primary Key
Customer_ID	NUMBER(10)	Foreign Key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL
Order_Status	VARCHAR2(20)	-
Shipping_Address	VARCHAR2(200)	-
Delivery_Date	DATE	-

2.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	NUMBER(10)	Primary Key
Order_ID	NUMBER(10)	Foreign Key
Product_ID	NUMBER(10)	Foreign Key
Quantity	NUMBER(10)	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Discount	NUMBER(10,2)	DEFAULT 0.00
Total_Price	NUMBER(10,2)	NOT NULL

3. Create Orders Table

```
SQL> CREATE TABLE Orders (  
2     Order_ID NUMBER(10) PRIMARY KEY,  
3     Customer_ID NUMBER(10) NOT NULL,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2) NOT NULL,  
6     Order_Status VARCHAR2(20),  
7     Shipping_Address VARCHAR2(200),  
8     Delivery_Date DATE,  
9     CONSTRAINT FK_ORDERS_CUSTOMER  
10         FOREIGN KEY (Customer_ID)  
11         REFERENCES Customer(Customer_ID)  
12 );
```

Table created.

4. Create Order_Details Table

```
SQL> CREATE TABLE Order_Details (  
2     Order_Detail_ID NUMBER(10) PRIMARY KEY,  
3     Order_ID NUMBER(10) NOT NULL,  
4     Product_ID NUMBER(10) NOT NULL,  
5     Quantity NUMBER(10) NOT NULL,  
6     Unit_Price NUMBER(10,2) NOT NULL,  
7     Discount NUMBER(10,2) DEFAULT 0,  
8     Total_Price NUMBER(10,2) NOT NULL,  
9     CONSTRAINT FK_ORDER_DETAILS_ORDER  
10         FOREIGN KEY (Order_ID)  
11         REFERENCES Orders(Order_ID)  
12 );
```

Table created.

5. Order Data Insertion

```
SQL> DESC Orders;  
Name                               Null?    Type  
-----  
ORDER_ID                           NOT NULL NUMBER(10)  
CUSTOMER_ID                        NOT NULL NUMBER(10)  
ORDER_DATE                         NOT NULL DATE  
TOTAL_AMOUNT                       NOT NULL NUMBER(10,2)  
ORDER_STATUS                       NOT NULL VARCHAR2(20)  
SHIPPING_ADDRESS                   NOT NULL VARCHAR2(200)  
DELIVERY_DATE                      NOT NULL DATE  
  
SQL> DESC Order_Details;  
Name                               Null?    Type  
-----  
ORDER_DETAIL_ID                    NOT NULL NUMBER(10)  
ORDER_ID                           NOT NULL NUMBER(10)  
PRODUCT_ID                        NOT NULL NUMBER(10)  
QUANTITY                           NOT NULL NUMBER(10)  
UNIT_PRICE                         NOT NULL NUMBER(10,2)  
DISCOUNT                         NOT NULL NUMBER(10,2)  
TOTAL_PRICE                        NOT NULL NUMBER(10,2)
```

```
SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(1, 'Aarav', 'Mehta', 'Male', DATE '2003-04-12', 'aarav.mehta@gmail.com', '9000000001', 'Adyar', 'Chennai', 'Tamil Nadu', '600020', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(2, 'Diya', 'Nair', 'Female', DATE '2004-07-25', 'diya.nair@gmail.com', '9000000002', 'Vyttila', 'Kochi', 'Kerala', '682015', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(3, 'Rohan', 'Verma', 'Male', DATE '2002-10-08', 'rohan.verma@gmail.com', '9000000003', 'Gauti Nagar', 'Lucknow', 'Uttar Pradesh', '226018', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(4, 'Kavya', 'Iyer', 'Female', DATE '2003-12-16', 'kavya.iyer@gmail.com', '9000000004', 'Mylapore', 'Chennai', 'Tamil Nadu', '600004', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(5, 'Arjun', 'Singh', 'Male', DATE '2004-01-29', 'arjun.singh@gmail.com', '9000000005', 'Sector 17', 'Chandigarh', 'Chandigarh', '160017', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(6, 'Nandhini', 'Krishnan', 'Female', DATE '2003-05-14', 'nandhini.krishnan@gmail.com', '9000000006', 'Gandhipuram', 'Coimbatore', 'Tamil Nadu', '641012', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(7, 'Vishal', 'Joshi', 'Male', DATE '2002-09-21', 'vishal.joshi@gmail.com', '9000000007', 'Malviya Nagar', 'Jaipur', 'Rajasthan', '302017', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(8, 'Pooja', 'Rao', 'Female', DATE '2004-03-06', 'pooja.rao@gmail.com', '9000000008', 'Kukatpally', 'Hyderabad', 'Telangana', '500072', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(9, 'Sanjay', 'Das', 'Male', DATE '2003-11-19', 'sanjay.das@gmail.com', '9000000009', 'Salt Lake', 'Kolkata', 'West Bengal', '700091', SYSDATE);
1 row created.

SQL> INSERT INTO Customer(Customer_ID, First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Address, City, State, Pincode, Registration_Date) VALUES(10, 'Harini', 'Shah', 'Female', DATE '2002-08-27', 'harini.shah@gmail.com', '9000000010', 'Vastrapur', 'Ahmedabad', 'Gujarat', '380015', SYSDATE);
1 row created.
```

6.Order_Details Data Insertion

```
SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(101, 1, DATE '2026-09-20', 790.00, 'Delivered', 'Chennai', DATE '2026-09-23');
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(102, 2, DATE '2026-09-21', 599.00, 'Delivered', 'Kochi', DATE '2026-09-24');
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(103, 3, DATE '2026-09-22', 999.00, 'Shipped', 'Bangalore', NULL);
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(104, 4, DATE '2026-09-23', 450.00, 'Delivered', 'Chennai', DATE '2026-09-26');
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(105, 5, DATE '2026-09-24', 750.00, 'Shipped', 'Mumbai', NULL);
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(106, 6, DATE '2026-09-25', 1200.00, 'Pending', 'Hyderabad', NULL);
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(107, 7, DATE '2026-09-26', 650.00, 'Delivered', 'Delhi', DATE '2026-09-29');
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(108, 8, DATE '2026-09-27', 899.00, 'Shipped', 'Pune', NULL);
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(109, 9, DATE '2026-09-28', 550.00, 'Pending', 'Kolkata', NULL);
1 row created.

SQL> INSERT INTO Orders(Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address, Delivery_Date) VALUES(110, 10, DATE '2026-09-29', 1100.00, 'Delivered', 'Ahmedabad', DATE '2026-10-02');
1 row created.
```

```
SQL> SELECT * FROM Orders
2 ORDER BY Order_ID;

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
101          1 20-SEP-26          790 Delivered
Chennai
23-SEP-26

102          2 21-SEP-26          599 Delivered
Kochi
24-SEP-26

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
103          3 22-SEP-26          999 Shipped
Bangalore

104          4 23-SEP-26          450 Delivered
Chennai

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
24-SEP-26

105          5 24-SEP-26          750 Shipped
Mumbai

106          6 25-SEP-26         1200 Pending

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
Hyderabad

107          7 26-SEP-26          650 Delivered
Delhi
29-SEP-26

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
Pune

108          8 27-SEP-26          899 Shipped

109          9 28-SEP-26          550 Pending
Kolkata

ORDER_ID CUSTOMER_ID ORDER_DATE TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----

110         10 29-SEP-26         1100 Delivered
Ahmedabad
02-OCT-26

10 rows selected.
```

7. Order Modification Operations

```
SQL> UPDATE Orders
  2  SET Order_Status = 'Delivered',
  3  Delivery_Date = DATE '2026-10-03'
  4  WHERE Order_ID = 106;

1 row updated.

SQL> SELECT * FROM Orders
  2  WHERE Order_ID = 106;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
      106           6 25-SEP-26          1200 Delivered
Hyderabad
03-OCT-26

SQL> COMMIT;

Commit complete.
```

8. Order_Details Modification

```
SQL> UPDATE Orders
  2  SET Order_Status = 'Shipped'
  3  WHERE Order_ID = 111;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL> UPDATE Orders
  2  SET Delivery_Date = DATE '2026-10-06'
  3  WHERE Order_ID = 111;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL> SELECT * FROM Orders
  2  WHERE Order_ID = 111;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-----
SHIPPING_ADDRESS
-----
DELIVERY_
-----
      111           1 03-OCT-26           850 Shipped
Chennai
06-OCT-26
```

9. Customer Order History Report

```

2  SELECT
3      c.Customer_ID,
4      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
5      o.Order_ID,
6      o.Order_Date,
7      od.Product_ID,
8      od.Quantity,
9      od.Unit_Price,
10     od.Total_Price,
11     o.Order_Status
12 FROM   Customer c
13 JOIN   Orders o
14       ON c.Customer_ID = o.Customer_ID
15 JOIN   Order_Details od
16       ON o.Order_ID = od.Order_ID
17 ORDER BY c.Customer_ID, o.Order_Date;

```

<u>CUSTOMER_ID</u>	<u>CUSTOMER_NAME</u>	<u>ORDER_ID</u>	<u>ORDER_DAT</u>	<u>PRODUCT_ID</u>	<u>QUANTITY</u>	<u>UNIT_PRICE</u>	<u>TOTAL_PRICE</u>
		<u>ORDER_STATUS</u>					
		1					
Karav	Mohit	181	20-SEP-26	1	2	399	798
	Delivered						

```

CUSTOMER_ID
CUSTOMER_NAME
-----
ORDER_ID ORDER_DATE PRODUCT_ID QUANTITY UNIT_PRICE TOTAL_PRICE
-----
ORDER STATUS
-----
2
Diya Hair 162 21-SEP-26 2 1 599 599
Delivered

```

```

CUSTOMER_ID    CUSTOMER_NAME
-----
ORDER_ID    ORDER_DATE    PRODUCT_ID    QUANTITY    UNIT_PRICE    TOTAL_PRICE
-----
ORDER_STATUS
-----
3
Rohan Verma
183 22-SEP-26      3      2      499.5      999
Shipped

```

CUSTOMER_ID						
CUSTOMER_NAME						
ORDER_ID	ORDER_DATE	PRODUCT_ID	QUANTITY	UNIT PRICE	TOTAL PRICE	
ORDER STATUS						
4						
Kavya Iyer	19-SEP-26	4	1	450	450	
Delivered						

```

CUSTOMER_ID _____
CUSTOMER_NAME _____
ORDER_ID ORDER_DAT PRODUCT_ID QUANTITY UNIT_PRICE TOTAL_PRICE
-----
ORDER_STATUS _____
6
Nandhini Krishnan
166 25-SEP-26 6 3 400 1200
Delivered

```

<u>CUSTOMER_ID</u>	<u>CUSTOMER_NAME</u>	<u>ORDER_ID</u>	<u>ORDER_DATE</u>	<u>PRODUCT_ID</u>	<u>QUANTITY</u>	<u>UNIT_PRICE</u>	<u>TOTAL_PRICE</u>
		<u>ORDER_STATUS</u>					
		7					
Vishal Joshi		187	26-SEP-26	7	1	650	650
Delivered							

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE
		ORDER_STATUS					
		8					
Pooja Rao		188	27-SEP-26	8	2	449.5	899
Shipped							

CUSTOMER_ID						
CUSTOMER_NAME						
ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE	
ORDER_STATUS						
9						
Sanjay Das	189	28-SEP-26	9	1	550	550
Pending						

CUSTOMER_ID						
CUSTOMER_NAME						
ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	TOTAL_PRICE	
ORDER_STATUS						
10						
Marini Shah						
110	29-SEP-26	10	2	550	1100	
Delivered						

10 rows selected.

10. Customer-wise Order Summary

```
SQL> SELECT
  2   c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3   o.Order_ID,
  4   o.Order_Date,
  5   o.Total_Amount,
  6   o.Order_Status
  7 FROM Customer c
  8 JOIN Orders o
  9 ON c.Customer_ID = o.Customer_ID
10 WHERE c.Customer_ID < 3
11 ORDER BY o.Order_Date;
```

CUSTOMER_NAME	ORDER_ID	ORDER_DATE	TOTAL_AMOUNT	ORDER_STATUS
Aarav Rohra	181	28-SEP-26	798	Delivered
Aarav Rohra	111	03-OCT-26	850	Shipped

```
SQL> SELECT
  2   c.Customer_ID,
  3   c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4   COUNT(o.Order_ID) AS Total_Orders,
  5   SUM(o.Total_Amount) AS Total_Spent
  6 FROM Customer c
  7 JOIN Orders o
  8 ON c.Customer_ID = o.Customer_ID
  9 GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
10 ORDER BY c.Customer_ID;
```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_SPENT
1	Aarav Rohra	2	1648
2	Elyse Nair	1	800

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_SPENT
3	Bhuvan Verma	1	999

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_SPENT
4	Kavya Zyer	1	400
5	Arjun Singh	1	798

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_SPENT
6	Ramdhani Krishnan	1	1200
7	Vishal Joshi	1	650

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_SPENT
8	Pooja Rao	1	300
9	Sanjay Das	1	550

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_SPENT
10	Marini Shah	1	1100

10 rows selected.
SQL> COMMIT;

11. Conclusion

The **Orders** and **Order_Details** tables were successfully designed and implemented for the **Vilvah Database Management System**. The tables store customer orders, product details, quantities, prices, discounts, and total amounts. Order insertion and modification operations were performed successfully. Customer order history and customer-wise order summary reports were also generated using SQL queries.